机电系统建模、辨识与控制
Modeling, Identification and Control of Mechatronic Systems

MIC_L7 SISO Design Limitations

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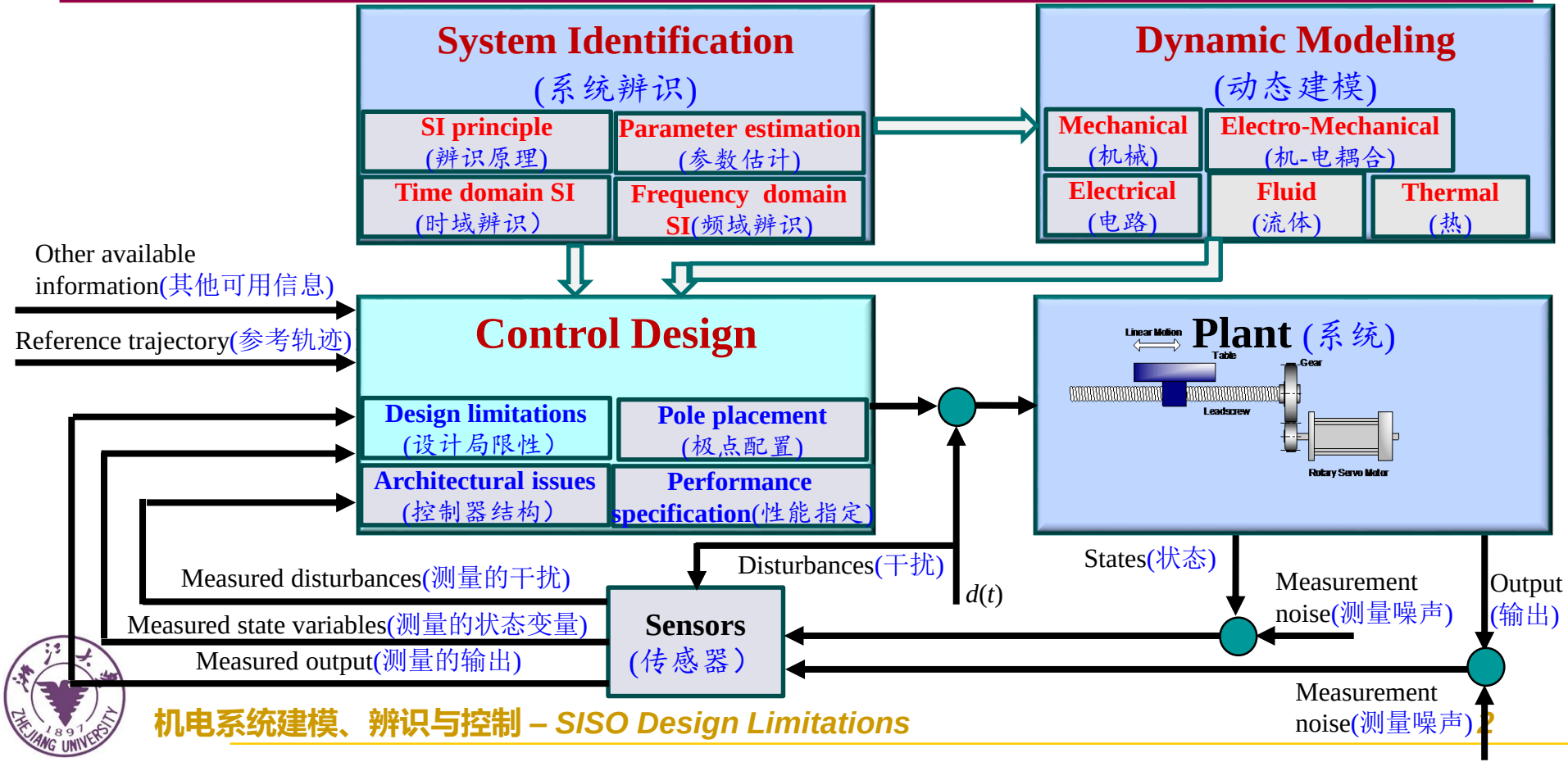
Main Contents of Course(课程内容)



Common Controller Design Issues

- **Closed-Loop Performance Specifications** （闭环性能指定）
  - **Command Following** （命令跟踪）
    - Response speed, Steady-state error , . . . （响应速度，稳态误差）
  - **Disturbance Attenuation Properties** （干扰抑制能力）
    - Rejection of disturbance at steady-state, transient speed, . . . （在稳态时对干扰的抑制，瞬态速度……）
  - **Practical Realizability** （可实现性）
    - Required control amplitude & control rate changes , . . . （所要求的控制幅度和控制量变化率）
  - **Transient Response Quality** （瞬态响应品质）
    - Overshoot, if the system is open-loop unstable （超调，如果系统有开环不稳定）
    - Undershoot, if the system is non-minimum phase （负调，如果系统是非最小相位）
  - **Sensitivity of Design to Modeling Errors** （对建模误差的设计敏感度）

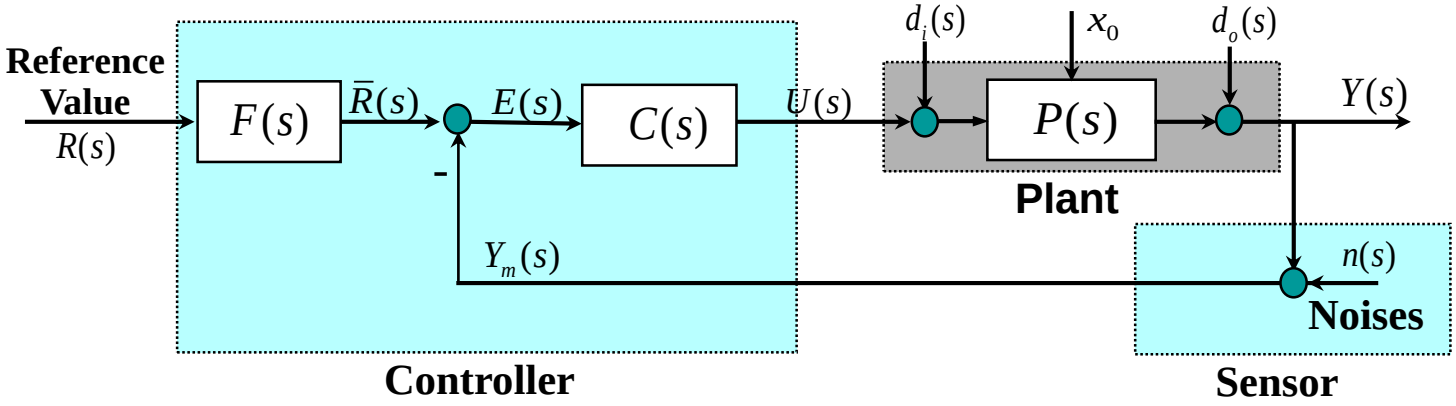
Fundamental SISO Design Trade-Offs for Bandwidth

- **Disturbance Rejection Property**

$$S(s)=\frac{1}{1+P(s)C(s)}$$

$$T(s)=\frac{P(s)C(s)}{1+P(s)C(s)}$$

$$S_i(s)=\frac{P(s)}{1+P(s)C(s)}$$



$$Y_{dis}(s)=S(s)\left[d_o(s)+P(s)d_i(s)\right]$$

Disturbances are rejected only at frequencies where

$$|S(j\omega)|\approx 0 \quad \text{or} \quad |T(j\omega)|\approx 1$$

For sinusoidal output disturbance

$$d_o(t)=A_d\sin(\omega t)$$

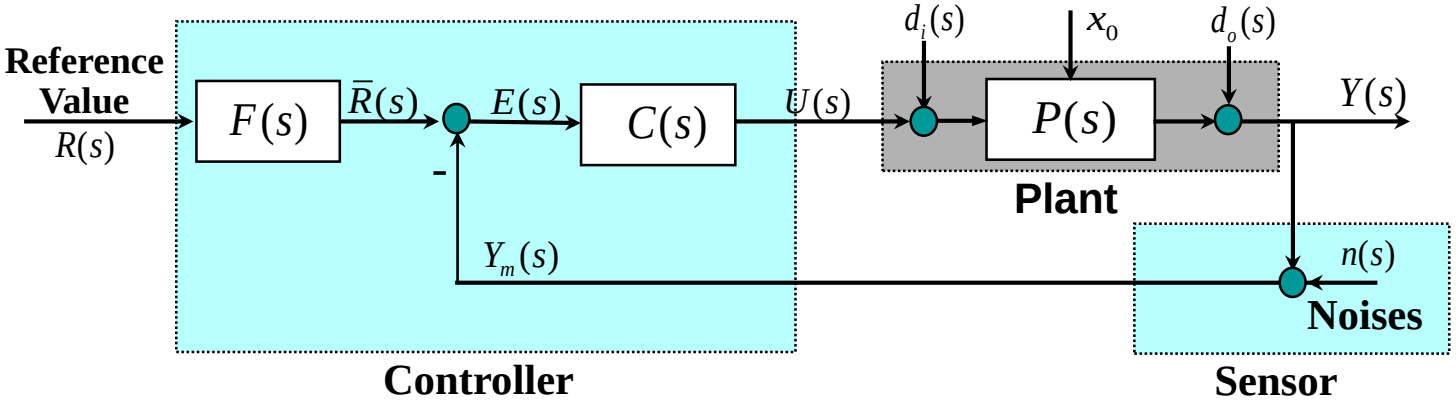
$$\Rightarrow y_{dis}(t)\to y_{ss}(t)=|S(j\omega)|\sin(\omega t+\angle S(j\omega))$$

To achieve acceptable performance in the presence of disturbances, it will generally be necessary to place a lower bound on the closed- loop bandwidth !

(为了在有干扰的情况下达到可接受的性能，通常需要设置闭环带宽的一个下界)

Fundamental SISO Design Trade-Offs for Bandwidth

- **Measurement Noise Effect**



$$S(s) = \frac{1}{1+P(s)C(s)}$$

$$T(s) = \frac{P(s)C(s)}{1+P(s)C(s)}$$

$$S_i(s) = \frac{P(s)}{1+P(s)C(s)}$$

$$Y_{measure}(s) = -T(s)n(s)$$

Measurement noise is rejected only at frequencies where

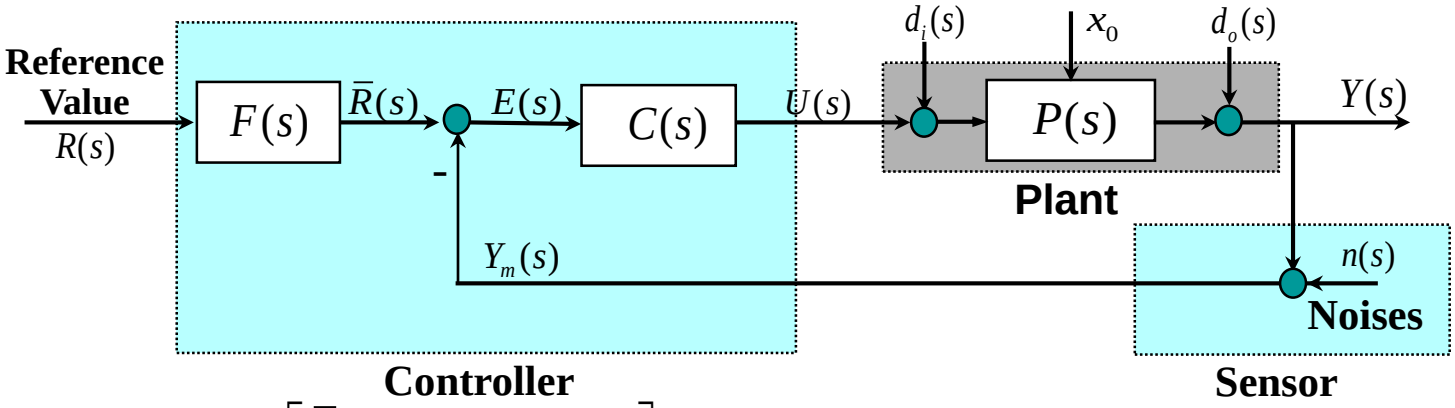
$$|T(j\omega)| \approx 0 \qquad |S(j\omega)| \approx 1$$

Given the fact that noise is typically dominated by high frequencies, there will be an upper limit on the bandwidth of closed-loop set by measurement noise !

(考虑到噪声通常由高频主导的事实，测量噪声会需要设置闭环带宽的一个上界)

Fundamental SISO Design Trade-Offs for Bandwidth

- **Control Effort**



$$S(s)=\frac{1}{1+P(s)C(s)}$$

$$T(s)=\frac{P(s)C(s)}{1+P(s)C(s)}$$

$$S_i(s)=\frac{P(s)}{1+P(s)C(s)}$$

$$S_u(s)=\frac{C(s)}{1+P(s)C(s)}=\frac{T(s)}{P(s)}$$

$$U(s)=S_u(s)\left[\bar{R}(s)-d_o(s)-n(s)\right]$$

For frequency range that $|S_u(j\omega)|>>1$ ⟶ Large control input may be needed at these frequencies

Large control signals arise at frequencies where $|T(j\omega)|\approx 1$ but $|P(j\omega)|<<1$ which occurs when closed-loop is forced to be much more responsive than open-loop process(在$|T(j\omega)|\approx 1$但$|P(j\omega)|<<1$的频段需要较大的控制信号，这也是闭环控制而不是开环控制起主导的时候)

To avoid actuator saturation or slew rate problems, it will generally be necessary to place an upper limit on the closed- loop bandwidth!

(为避免执行器饱和或其变化率过大问题，通常有必要为闭环带宽设置一个上界)

Fundamental SISO Design Trade-Offs for Bandwidth

- **Modeling Error Effect**

Robust Stability Theorem: $\left|T_n(j\omega)\right|\left|P_\Delta(j\omega)\right| < 1, \quad \forall\omega$

Design Constraint: $S_n(j\omega) + T_n(j\omega) = 1, \quad \forall\omega$

$\Longrightarrow$ For frequencies with significant modeling errors $|P_\Delta(j\omega)|>>1$

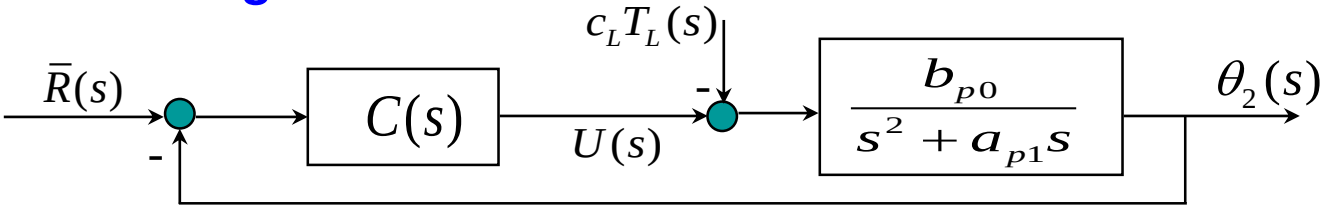
$$\Rightarrow \left|T_n(j\omega)\right| << 1 \text{ or } \left|S_n(j\omega)\right| \approx 1$$

Being responsive to reference changes and against disturbances at frequencies with significant modeling errors jeopardize stability; note also that MME $|P_\Delta(j\omega)|$ is normally significant at high frequencies! (对参考量变化有响应和在有显著建模误差的频段上的抗干扰，会危及稳定性；注意MME $|P_\Delta(j\omega)|$通常在高频段比较显著)

To achieve robust stability and acceptable performance in the presence of modeling errors, it will generally be desirable to place an upper limit on closed-loop bandwidth !
(为获得鲁棒稳定性和有建模误差存在下可接受的性能，通常希望为闭环带宽设置一个上界)

Case Study of Bandwidth Selection

- **PID Control of Rigid Belt ECP Emulator**



A proper PID controller has been previously synthesized to place CL poles at

$$C(s) = K_P + \frac{K_I}{s} + \frac{K_D s}{\tau_D s + 1} = \frac{(K_D + K_P\tau_D)s^2 + (K_P + K_I\tau_D)s + K_I}{\tau_D s^2 + s}$$

PID 1 $$C_1(s) = \frac{b_{c2}s^2 + b_{c1}s + b_{c0}}{s^2 + a_{c1}s} = \frac{8.33s^2 + 205s + 2054}{s^2 + 117s}$$

$p_{CL1,2d} = -20 \pm 20j,\ p_{CL3,4d} = -40$
$\omega_n = 28\text{rad/s}, \zeta = 0.707$

PID 2 $$C_2(s) = \frac{12.39s^2 + 385s + 4620.6}{s(s + 136.5)}$$ or $K_p = 2.57$ $K_I = 33.85$ $K_d = 0.072$ $\tau_D = 0.0073$

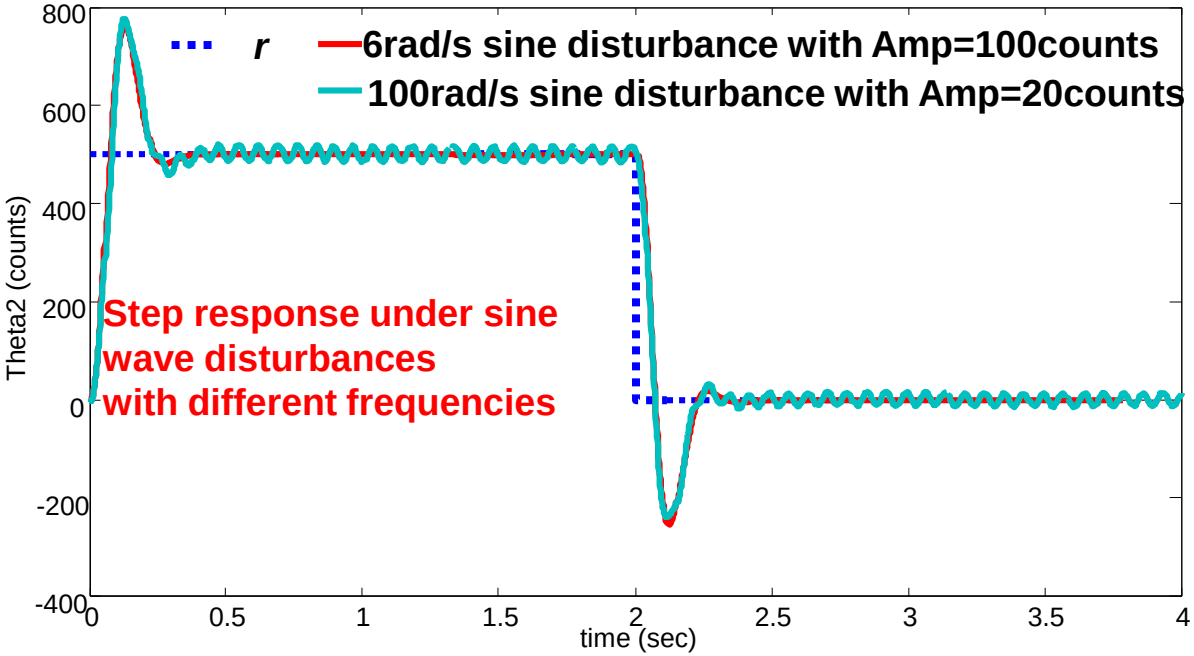
$\omega_n = 30\sqrt{2} = 42\text{rad/s}, \zeta = 0.707$
$p_{CL1,2d} = -30 \pm 30j,\ p_{CL3,4d} = -40$

PID 2 $$C_3(s) = \frac{34.45s^2 + 1550s + 26729}{s(s + 236.5)}$$ or $K_p = 6.075$ $K_I = 113$ $K_d = 0.12$ $\tau_D = 0.004$

$\omega_n = 58\text{rad/s}, \zeta = 0.857,$
$p_{CL1,2d} = -50 \pm 30j,\ p_{CL3,4d} = -70$

Case Study of Bandwidth Selection

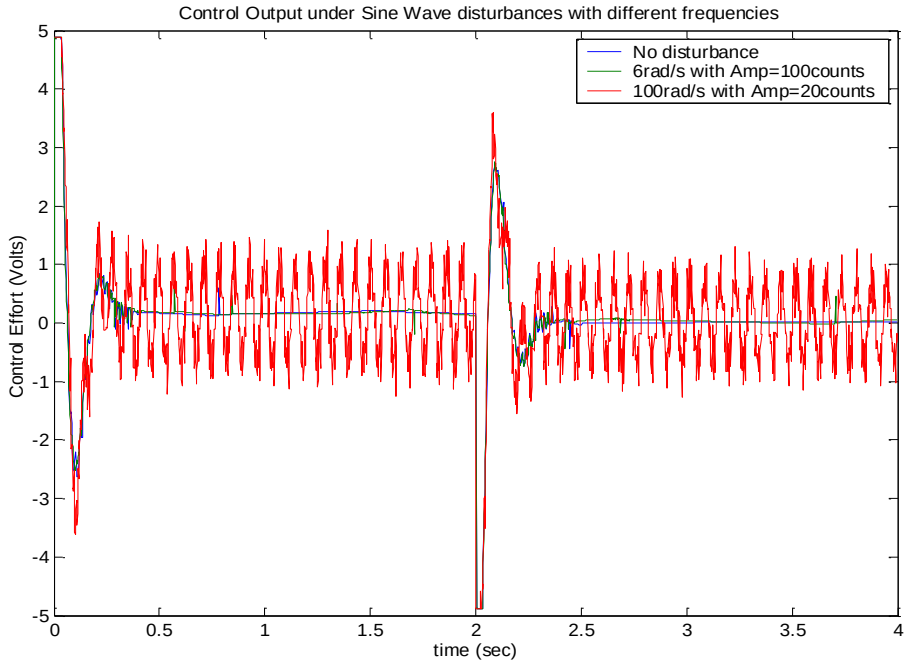
- **Experimental Step Response under Sine Wave Disturbances with PID1**



- For disturbances within the CL bandwidth (e.g., 6 rad/sec), the system has very good disturbance rejection capability due to low sensitivity. (6rad/s的正弦干扰，作用在干扰电机上的电压是0.065V,等效于100个脉冲)
- For disturbances outside the CL bandwidth (e.g., 100 rad/sec), due to the low complementary sensitivity or sensitivity value close to 1, the effect of disturbance is not attenuated. (对于100rad/s的干扰，作用在干扰电机上的电压是3.14V，等效于20个脉冲)

Case Study of Bandwidth Selection

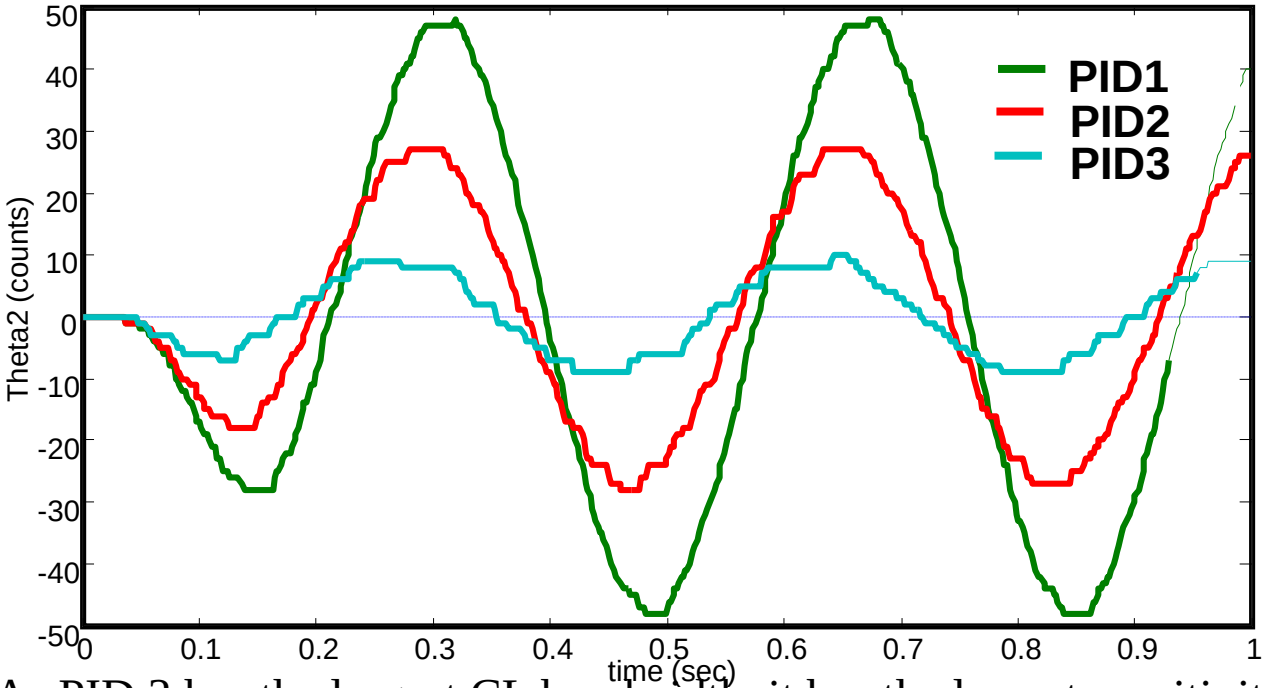
- **Experimental Control Effort Sine Wave Disturbances with PID1**



The large amplitude of sinusoidal control input at steady-state for sinusoidal output disturbance of 100 rad/sec is due to the large value of the control input sensitivity function at 100 rad/sec. (由于控制输入灵敏度函数值在100 rad/sec时较大，所以100 rad/sec的正弦输出干扰下的稳态时控制输入的幅值较大)

Case Study of Bandwidth Selection

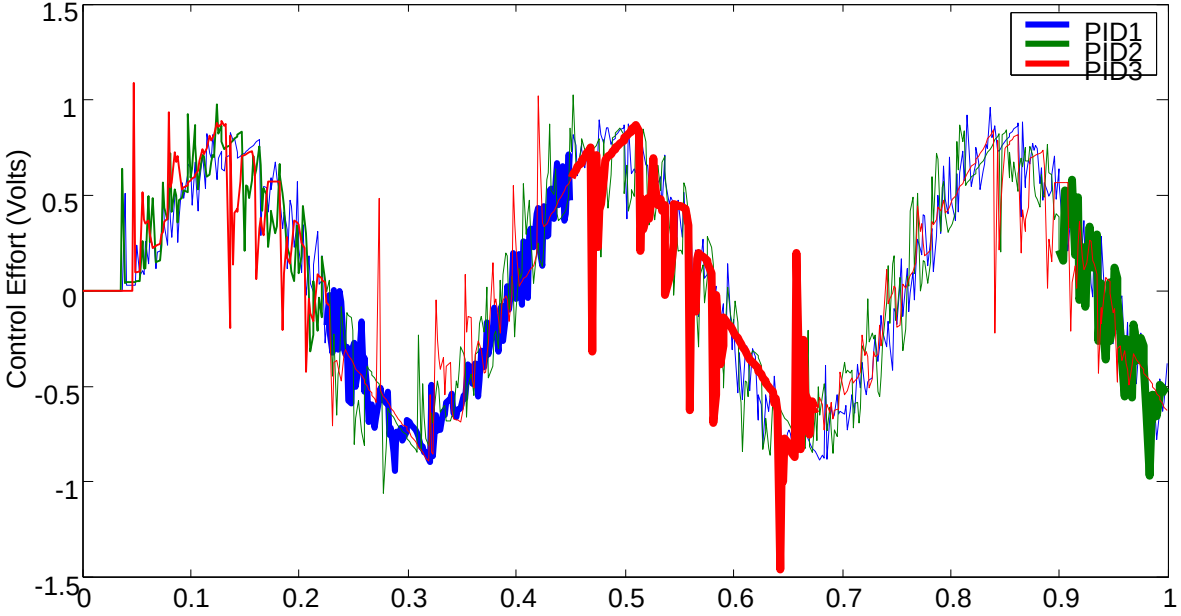
- **Response under Sine Wave disturbance of 20 rad/sec with Amp = 200 counts**



As PID 3 has the largest CL bandwidth, it has the lowest sensitivity value at 20 rad/sec. As such, it has the best disturbance rejection capability. (PID 3的闭环带宽最大，在ω= 20rad/sec的灵敏度值最低。因而具有最佳的抗干扰能力)

Case Study of Bandwidth Selection

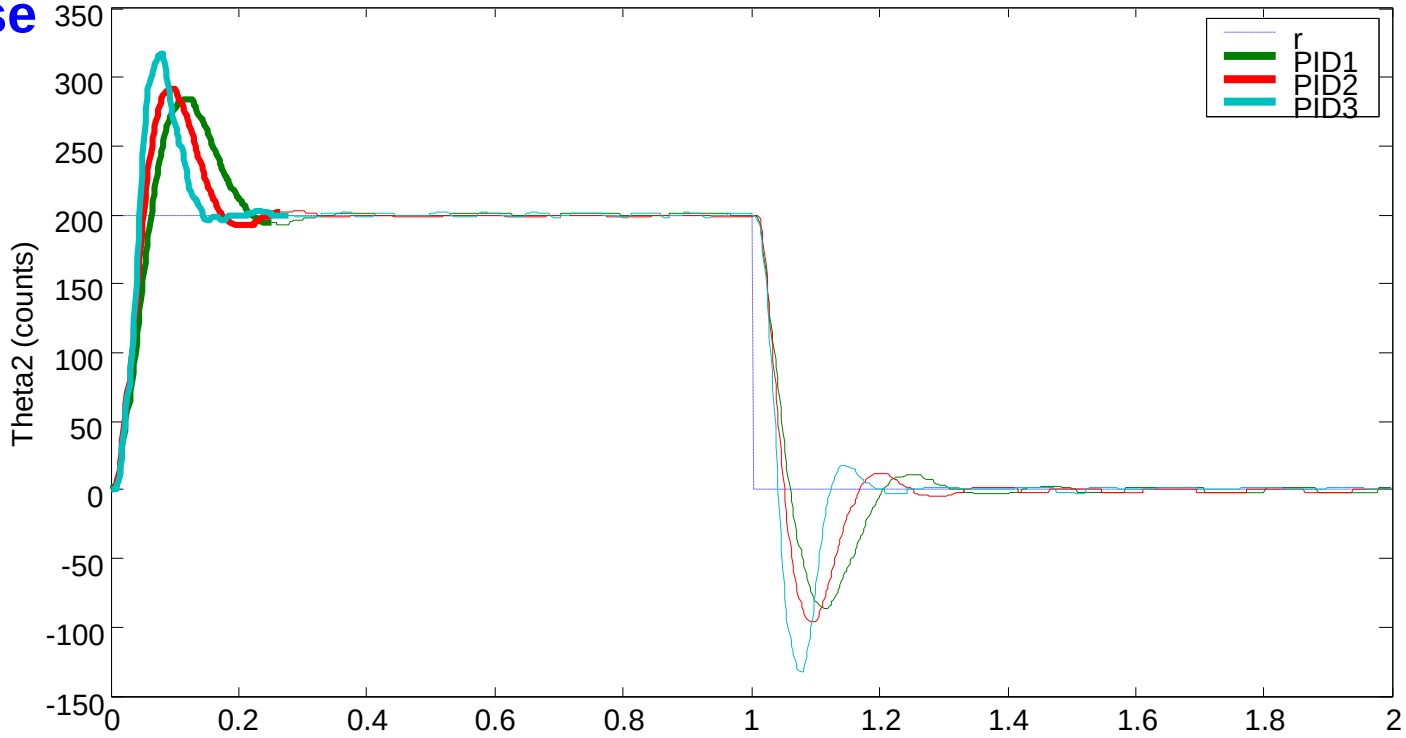
- **Control Output under Sine Wave disturbance of 20 rad/sec with Amp = 200 counts**



As PID 3 has the largest CL bandwidth, it has the highest control input sensitivity value at all frequencies. As such, its control input tends to be choppier than PID1 and PID2 (由于PID 3具有最大的闭环带宽,它在所有频段都有最高的控制输入灵敏度值。因此,它的控制量往往比PID1和PID2更波动)

Case Study of Bandwidth Selection

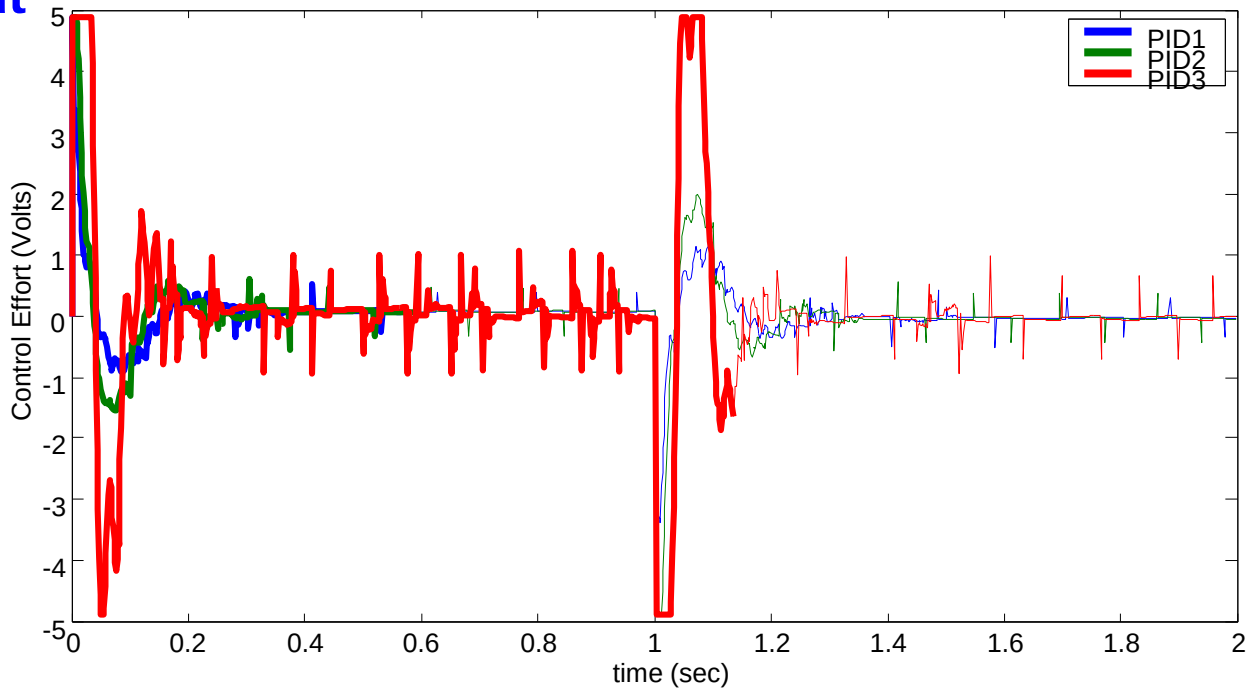
- **Step Response**



As PID 3 has the largest CL bandwidth, it has the fastest response time (由于PID 3的闭环带宽最大，所以它的响应时间最快).

Case Study of Bandwidth Selection

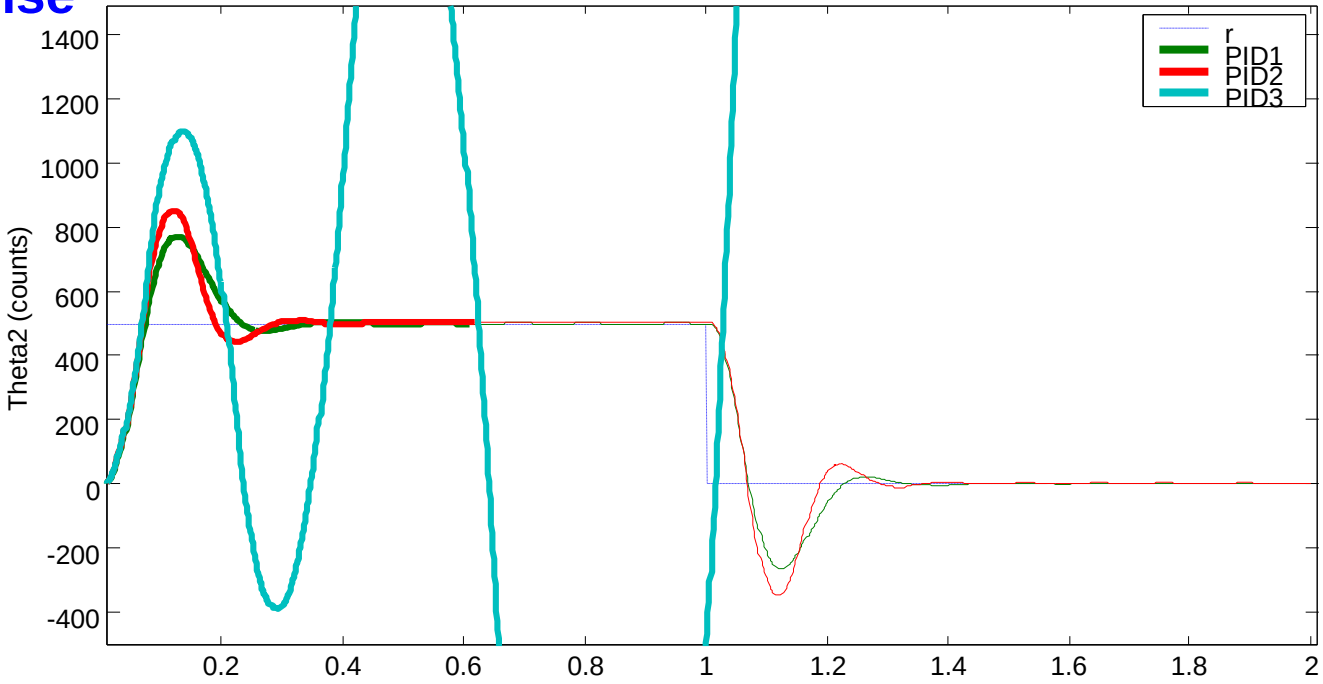
- **Control Input**



However, PID 3 is more prone to control input saturation and has higher control input sensitivity (choppy control input) (然而，PID 3更倾向于达到控制输入饱和且具有更高的控制输入灵敏度(波动的控制输入))

Case Study of Bandwidth Selection

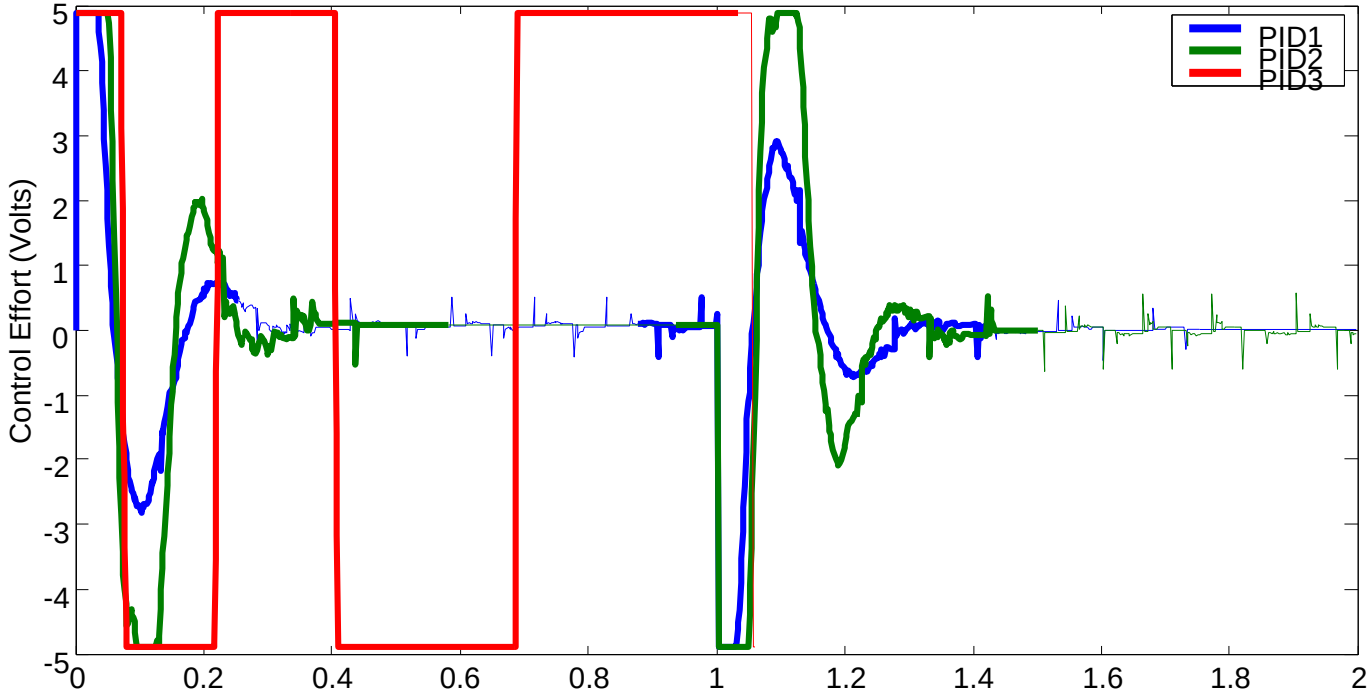
- **Step Response**



In fact, for large reference input command, PID 3 has such a severe control input saturation problem that actually results in an unstable CL system (实际上，对于大的参考输入命令，PID 3存在严重的控制输入饱和问题，导致闭环系统不稳定)

Case Study of Bandwidth Selection

- **Control Input**



The control input saturation problem of PID 3 and chopper control input of PID 2 when compared to PID 1 (与PID 1对比，PID 3的控制输入饱和PID 2的控制振荡问题)

Effect of Open Loop Integrators

- **Lemma 1 (System Types for <u>Reference Command</u>)**

For *stable* CL system with one-DOF controller configuration, assume that open-loop TF has i free integrators or i poles at the origin, i.e., (对采用单自由度控制器的稳定闭环系统，假设开环TF在原点处有i个自由积分器或i个极点，即)

$$D_p(s)D_c(s) = s^i \bar{D}_p(s)\bar{D}_c(s), \quad i \geq 1$$

$$\bar{D}_p(0)\bar{D}_c(0) = c_0 \neq 0$$

$$N_p(0)N_c(0) = c_1 \neq 0$$

Then, for a step output disturbance or step reference command, the error $e(t)$ satisfies (对于阶跃输出干扰或阶跃参考命令，误差$e(t)$满足)

$$\lim_{t\to\infty} e(t) = 0, \quad \forall i \geq 1$$

F.V.T

$$\int_0^\infty e(t)dt = 0, \quad \forall i \geq 2$$

$$E(s) = [1 - T(s)F(s)]R(s) - S_i(s)d_i(s) - S(s)d_0(s) + T(s)n(s)$$

Effect of Open Loop Integrators

- **Lemma 1 (cts)**

  Also, for a negative unit ramp output disturbance or a positive unit ramp reference command, the error $e(t)$ satisfies (对于负单位斜坡输出干扰或正单位斜坡参考命令，误差$e(t)$满足)

$$\lim_{t\to\infty} e(t) = \frac{c_0}{c_1}, \qquad \text{for} \quad i = 1$$

$$\lim_{t\to\infty} e(t) = 0, \qquad \forall i \geq 2$$

$$\int_0^\infty e(t)dt = 0, \qquad \forall i \geq 3$$

$$E(s) = [1 - T(s)F(s)]R(s) - S_i(s)d_i(s) - S(s)d_0(s) + T(s)n(s)$$

Effect of Open Loop Integrators

- **Lemma 2 (System Types for Input Disturbance)**

For *stable* CL system with one-DOF controller configuration, assume that the controller TF has i free integrators or i poles at the origin, i.e., (对于采用单自由度控制器的稳定闭环系统，假设控制器传递函数在原点处有i个自由积分器或i个极点，即)

$$D_c(s) = s^i \bar{D}_c(s), \quad \bar{D}_c(0) = c_0 \neq 0, \quad N_c(0) = c_1 \neq 0$$

Then, for a step input disturbance, the error $e(t)$ satisfies (对于阶跃输入干扰，误差$e(t)$满足)

$$\lim_{t\to\infty} e(t) = 0, \qquad \forall i \geq 1$$

$$\int_0^\infty e(t)dt = 0, \qquad \forall i \geq 2$$

Also, for a negative unit ramp input disturbance, the error $e(t)$ satisfies (对于负单位斜坡输入干扰，误差$e(t)$满足)

$$\lim_{t\to\infty} e(t) = \frac{c_0}{c_1}, \qquad \text{for} \quad i = 1$$

$$\lim_{t\to\infty} e(t) = 0, \qquad \forall i \geq 2$$

$$\int_0^\infty e(t)dt = 0, \qquad \forall i \geq 3$$

General Effect of Open Loop Poles/Zeros

- **Lemma 3**

For *stable* CL system with poles located to the left of $-\alpha$ for some $\alpha>0$, assume that the <u>controller has at least one pole at origin</u>. Then, for any uncancelled plant zero z_0 or pole p_0 that is to the right of CL poles (i.e., $R\{z_0\}>-\alpha$ or $R\{p_0\}>-\alpha$), we have （对于极点在$-\alpha(\alpha>0)$左侧的稳定闭环系统，假设控制器在原点处至少有一个极点。那么，对任何闭环极点右侧未消掉的系统零点或极点(也就是$R\{z_0\}>-\alpha$ 或$R\{p_0\}>-\alpha$)，我们有）

(i) For a positive unit reference step or a negative unit step output disturbance, error $e(t)$ satisfies （对于正单位参考阶跃或负单位阶跃输出干扰，误差$e(t)$满足）

$$E(s)=S(s)\underbrace{[R(s)-d_0(s)]}_{1/s}=S(s)\frac{1}{s}$$

At $z_0, S(z_0)=1$ (zeros of L(s))

$$\int_0^\infty e(t)e^{-z_0 t}dt=\frac{1}{z_0}$$

$$\int_0^\infty e(t)e^{-p_0 t}dt=0$$

According to definition of Laplase transform (根据拉普拉斯变换的定义)

$$\int_0^\infty e(t)e^{-z_0 t}dt=E(z_0)=S(z_0)\frac{1}{z_0}=\frac{1}{z_0}$$

At $p_0, S(p_0)=0$ (poles of L(s))

$$\int_0^\infty e(t)e^{-p_0 t}dt=E(p_0)=0$$

General Effect of Open Loop Poles/Zeros

(ii) For a positive unit step reference and for z_0 in RHP (unstable zero), (对于正的单位阶跃参考和右半平面中的z_0(不稳定零点)),

$$\int_0^\infty y(t)e^{-z_0 t}dt = 0$$

(iii) For a negative unit step input disturbance, (对于负的单位阶跃输入干扰)

$$\int_0^\infty e(t)e^{-z_0 t}dt = 0$$

$$\int_0^\infty e(t)e^{-p_0 t}dt = \frac{D_c(p_0)}{p_0 N_c(p_0)}$$

Observations for Effect of Open Loop Poles/Zeros

(i) A real *stable* (LHP) *zero* to the right of all CL poles produces *overshoot* in the step response（在所有闭环极点右侧的实稳定零点(LHP)会在阶跃响应中产生超调）

$$\left.\begin{array}{l}\Rightarrow \int_0^{\infty} e(t)e^{\alpha t}dt = -\dfrac{1}{\alpha} < 0 \\ \text{since } e(t) > 0 \text{ at least at the beginning}\end{array}\right\} \Rightarrow$$

$e(t)$ has to negative during time intervals!
($e(t)$在这段时间内必须为负)

$$z_0 = -\alpha, \alpha > 0$$

Small stable zero leads to large overshoot!
(小的稳定零点会导致大的超调)

(ii) A real *unstable* (RHP) *zero* always produces *undershoot* in the step response. The amount of undershoot grows as the zero approaches origin（实不稳定(RHP)零点总是在阶跃响应中产生负调。当零点接近原点时，负调量增加）

$$\left.\begin{array}{l}\int_0^{\infty} y(t)e^{-\alpha t}dt = 0, \quad \text{since } y(t) = 1 \text{ for } \forall t > t_s \\ \int_0^{t_s} y(t)e^{-\alpha t}dt + \int_{t_s}^{\infty} e^{-\alpha t}dt = 0, \Rightarrow \int_0^{t_s} y(t)e^{-\alpha t}dt + \dfrac{1}{\alpha}e^{-\alpha t_s} = 0\end{array}\right\} \Rightarrow$$

$y(t)$ has to negative during time intervals when $t<t_s$!（在$t<t_s$这段时间内必须为负）

$$z_0 = \alpha, \alpha > 0$$

Small unstable zero leads to large undershoot!
（小的不稳定零点会导致大的负调）

Observations for Effect of Open Loop Poles/Zeros

(iii) Any *real* open-loop *pole* to the right of all CL poles will produce *overshoot* in a one-DOF control architecture（在单自由度控制结构中，所有闭环极点右侧的任何实开环极点都会产生超调）

$$\int_0^{\infty} e(t)e^{-p_0 t}dt = 0$$

$\Rightarrow$ For fast unstable open-loop poles:（对快速不稳定开环极点来说）: $p_0 = \alpha,\ \alpha >> 1$

$e^{-p_0 t}$ drops zero very quickly. . ($e^{-p_0 t}$ 迅速降为0)

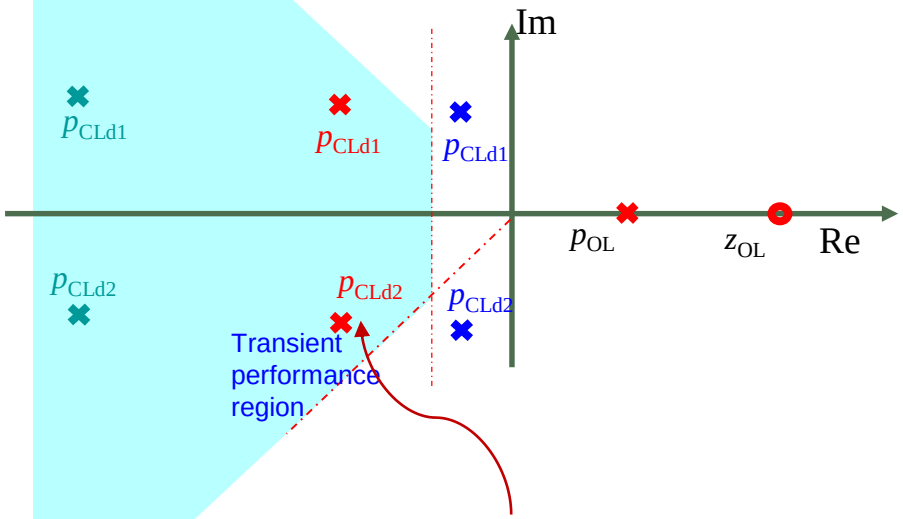
SISO Design Guidelines

- The *CL bandwidth* should be in practice be set *less* than the *smallest unstable zero* to avoid large undershoot in step response (在实际应用中，闭环带宽应小于最小的不稳定零点，以避免阶跃响应中出现较大的负调)
- It is advisable to set CL bandwidth *greater* than *real part* of any *unstable pole* to avoid large overshoot (建议闭环带宽大于任何不稳定极点的实部，以避免较大的超调)
- To avoid overshoot produced by *stable plant or controller zeros*, cancel them outside the loop in a two-DOF design using poles of filter TF $F(s)$ (为了避免系统或控制器的稳定零点产生的超调，在双自由度设计的闭环回路之外设计滤波器, 其传递函数是$F(s)$,用$F(s)$的极点来消去稳定零点)

SISO Design Guidelines

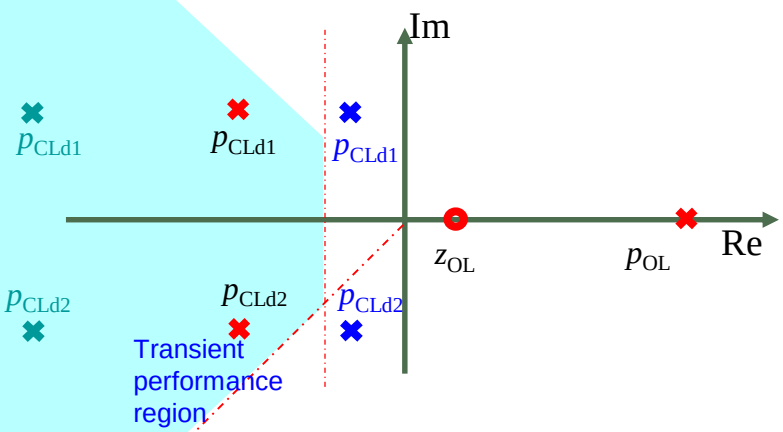
How to place the desired closed loop poles p_{CLd} for the following two cases?

Case 1



Good!

Case 2

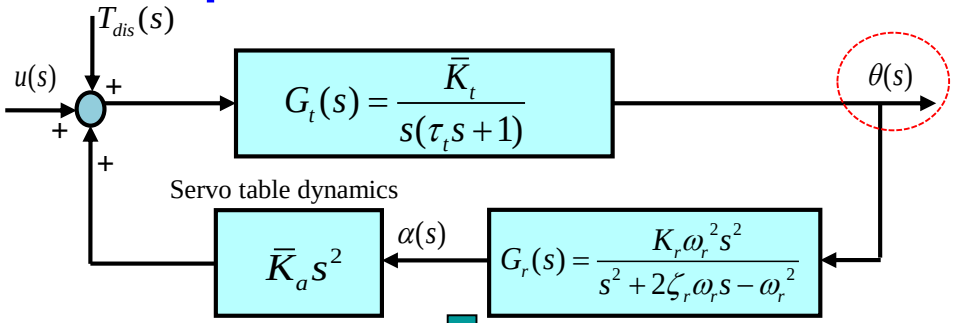


Is it satisfactory for such a pole placement?

NO!

Output Stabilization of Inverted Pendulum

- **Output Stabilization via Table Measurement Only**

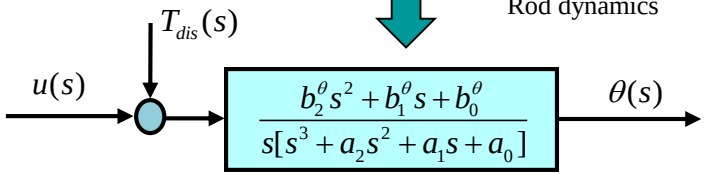


$$\theta(s)=[u(s)+T_{dis}(s)+\bar{K}_a s^2\frac{K_r\omega_r^2 s^2}{s^2+2\zeta_r\omega_r s-\omega_r^2}\theta(s)]\cdot\frac{\bar{K}_t}{s(\tau_t s+1)}\cdot$$

$$b_0^\theta=-\frac{K_t\omega_t\omega_r^2}{1-\bar{K}_a\bar{K}_t K_r\omega_t\omega_r^2},$$

$$b_1^\theta=\frac{2\zeta_r\omega_t\omega_r K_t}{1-\bar{K}_a\bar{K}_t K_r\omega_t\omega_r^2},$$

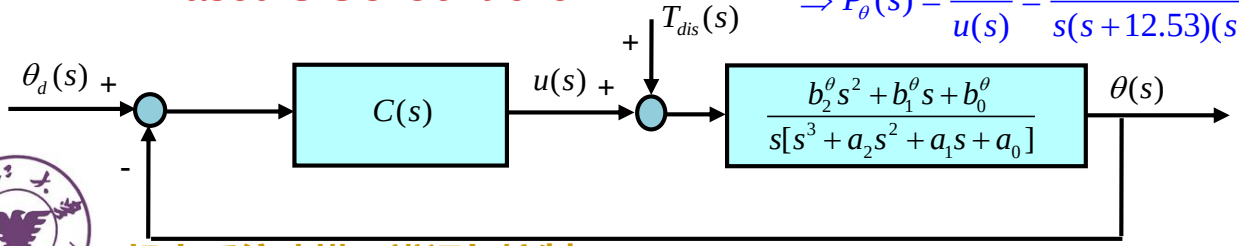
$$b_2^\theta=\frac{K_t\omega_t}{1-\bar{K}_a\bar{K}_t K_r\omega_t\omega_r^2},$$



TF Based SISO Controller

$$b_2^\theta s^2+b_1^\theta s+b_0^\theta K_t=10.36s^2+5.172s-452=10.357(s+6.8603)(s-6.3609)$$

$$\Rightarrow P_\theta(s)=\frac{\theta(s)}{u(s)}=\frac{10.357(s+6.8603)(s-6.3609)}{s(s+12.53)(s+0.7393)(s-9.8)}$$



Stabilization via Table Angle Measurement only

- **Controller Structure via Pole-Placement Design**

$$P_\theta(s) = \frac{10.357(s+6.8603)(s-6.3609)}{s(s+12.53)(s+0.7393)(s-9.8)} \qquad C(s) = \frac{b_{c4}s^4 + b_{c3}s^3 + b_{c2}s^2 + b_{c1}s + b_{c0}}{s(s^3 + a_{c2}s + a_{c1}s + a_{c0})}$$

we can arbitrarily place all CL poles: $p_{1,2d}$, p_{3d},p_{4d} p_{5d}, p_{6d},p_{7d},p_{8d}

$$A_{CL}(s) = D_p(s)D_c(s) + N_p(s)N_c(s)$$
$$= 10.357(s+6.8603)(s-6.3609)(b_{c4}s^4 + b_{c3}s^3 + b_{c2}s^2 + b_{c1}s + b_{c0})$$
$$+ s(s+12.53)(s+0.7393)(s-9.8)s(s^3 + a_{c2}s + a_{c1}s + a_{c0})$$
$$\Leftrightarrow A_{CLd}(s) = A_{CLd}(s) = \prod_{i=1}^{8}(s - p_{id})$$

Stabilization via Table Angle Measurement only

- **Rotary Inverted Pendulum**

$$P_\theta(s) = \frac{10.357(s+6.86)\boxed{(s-6.361)}}{s(s+12.53)(s+0.7393)\boxed{(s-9.8)}}$$

CL Poles

ω_{nd}=13.6, ζ_d=0.59;

-8.0240 + 10.9809i
-8.0240 – 10.9807i
-41.3749 + 0.0000i
-40.7014 + 1.0856i
-40.7014 - 1.0856i
-39.4944 + 1.0092i
-39.4944 - 1.0092i
-38.9534 + 0.0000i

$$\Rightarrow C(s) = \frac{1.339\times10^6 s^4 + 1.273\times10^7 s^3 - 7.336\times10^7 s^2 - 4.228\times10^8 s - 1.707\times10^9}{s^4 + 253.3s^3 - 1.384\times10^7 s^2 - 8.888\times10^7 s}$$

$$= \frac{1.3864\times10^7 (s+12.52)(s-7.281)(s+2.13+j3.07)(s+2.13-j3.07)}{s(s+3845.5)(s-3598.6)(s+6.4226)}$$

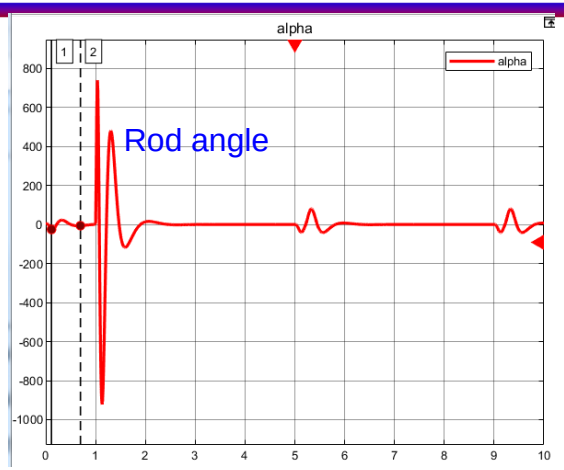
- **The closeness of the unstable plant pole at 9.8 and the non-minimum-phase plant zero at 6.361 indicates a very difficult control problem to solve in practice;**
- **Any output feedback stabilizing controller for these types of systems will lead to a CL system having unavoidable violent transient responses, which makes hardware implementation of the controller questionable.**

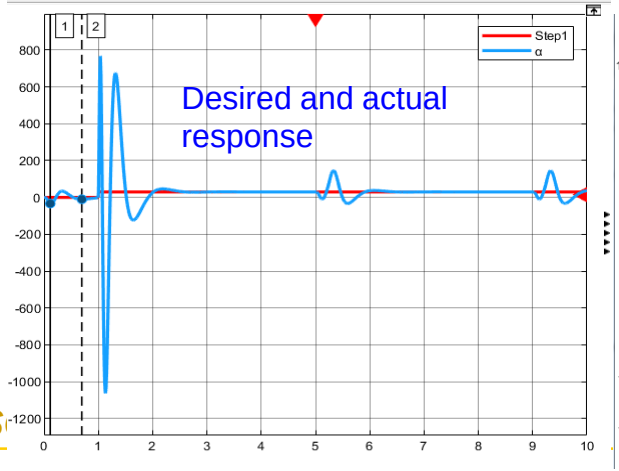
Stabilization via Table Angle Measurement only

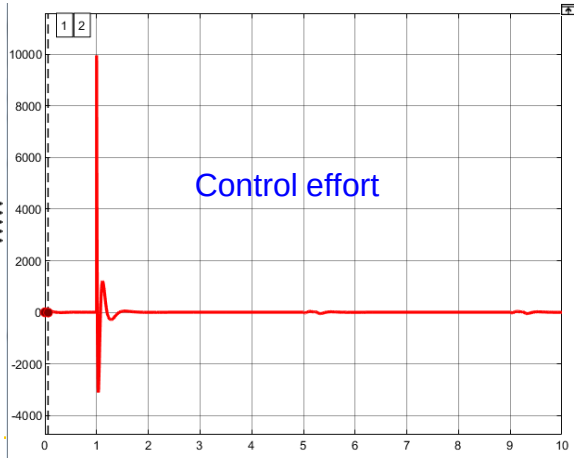
- **Simulation Results**

➢ **Although can be stable theoretically, the huge transient response leads to unimplementable control input and the violation of the small angles assumption used in the modeling process.**

➢ **The controller cannot be implemented on hardware !**

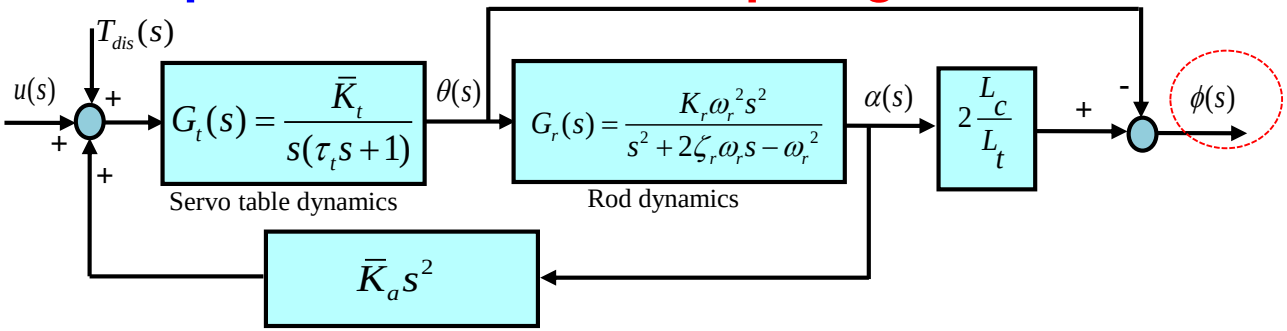






Output Stabilization of Inverted Pendulum

- **Output Stabilization via Tip Angle Measurement Only**

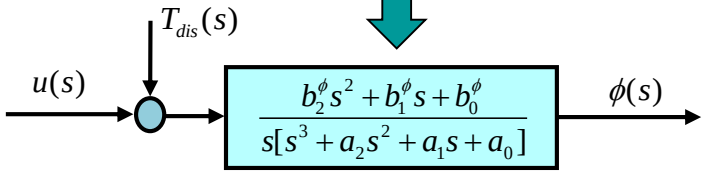


$$\phi(s) = 2\frac{L_c}{L_t}\alpha(s) - \theta(s)$$

$$b_0^\phi = 2\frac{L_c}{L_t}b_0^\alpha - b_0^\theta,$$

$$b_1^\phi = 2\frac{L_c}{L_t}b_1^\alpha - b_1^\theta,$$

$$b_2^\phi = 2\frac{L_c}{L_t}b_1^\alpha - b_2^\theta,$$

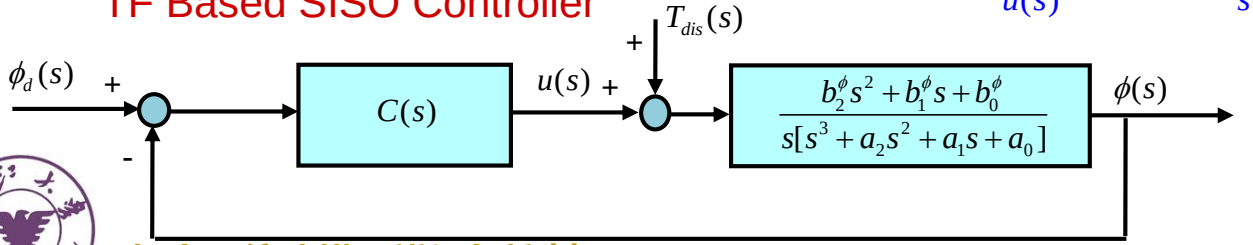


$$b_2^\phi s^2 + b_1^\phi s + b_0^\phi K_t = 5.1848s^2 - 5.172s + 452$$

$$= 5.1848(s - 0.4988 + j9.3232)(s - 0.4988 - j9.3232)$$

$$\Rightarrow P_\phi(s) = \frac{\phi(s)}{u(s)} = \frac{5.1848(s - 0.4988 + j9.3232)(s - 0.4988 - j9.3232)}{s(s+12.53)(s+0.7393)(s-9.8)}$$

TF Based SISO Controller



Stabilization via Tip Angle Measurement only

- **Rotary Inverted Pendulum**

$$P_\phi(s) = \frac{5.1848\boxed{(s-0.4988+j9.3232)}\boxed{(s-0.4988-j9.3232)}}{s(s+12.53)(s+0.7393)\boxed{(s-9.8)}}$$

$$C(s) = \frac{b_{c4}s^4 + b_{c3}s^3 + b_{c2}s^2 + b_{c1}s + b_{c0}}{s(s^3 + a_{c2}s + a_{c1}s + a_{c0})}$$

ω_{nd}=13.6,
ζ_d=0.59;

CL Poles

-8.0240 + 10.9809i
-8.0240 – 10.9807i
-42.1908 + 0.0000i
-41.0893 + 1.7918i
-41.0893 - 1.7918i
-39.0871 + 1.6818i
-39.0871 - 1.6818i
-38.1765 + 0.0000i

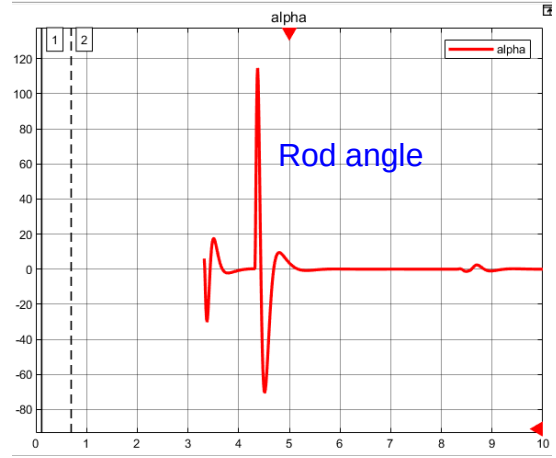
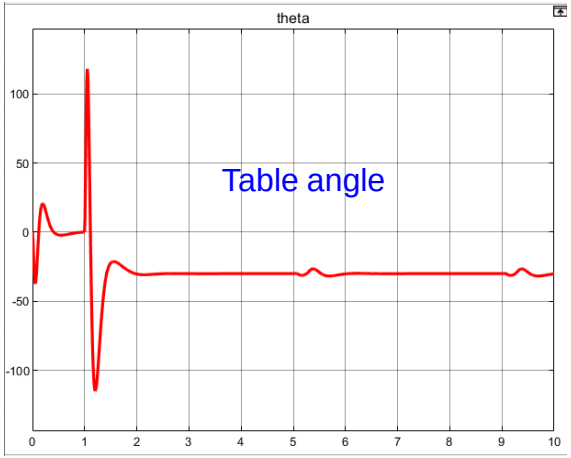
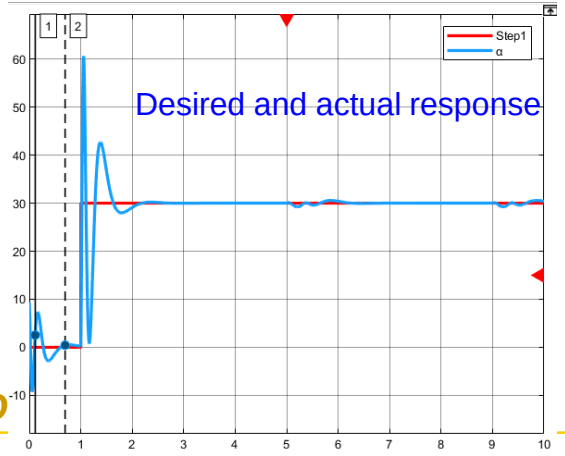
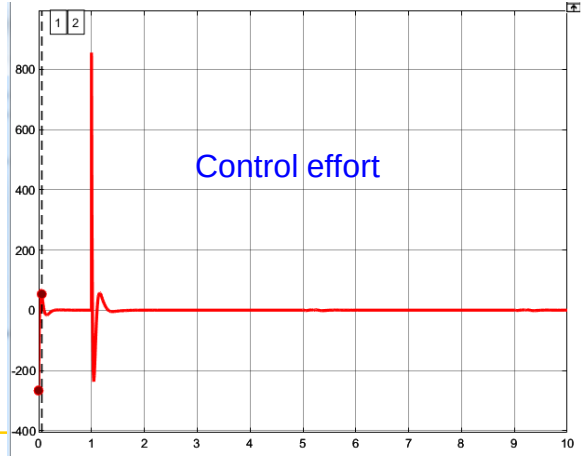
$$\Rightarrow C(s) = \frac{-1.049\times10^5 s^4 - 5.354\times10^5 s^3 + 3.294\times10^7 s^2 + 4.228\times10^8 s + 1.707\times10^9}{s^4 + 253.3s^3 + 5.712\times10^5 s^2 + 2.002\times10^6 s}$$

$$= \frac{8460(s-20.912)(s+12.792)(s+6.6123+j4.136)(s+6.6123-j4.136)}{s(s+124.89+j744.83)(s+124.89-j744.83)(s+3.5095)}$$

- **The separation of the unstable plant pole at 9.8 and the non-minimum-phase plant zero at 0.4988+j9.3232 is larger than that in table angle feedback case, indicating a relatively easier control problem to solve in practice.**

Stabilization via Tip Angle Measurement only

- **Simulation Results**

➢ **Still very large transient responses but less than those with table angle measurement case! This is due to the increased separation of the unstable plant pole and the non-minimum phase zero.**

➢ **Hardware implementation is still difficult due to large transient control effort.**